Akshay Patil

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EDUCATION

Georgia Institute of Technology – Atlanta, GA **August 2026 – May 2029**
Bachelor of Science in Aerospace Engineering
Current coursework: AE 1601 Aerospace Engineering, COE 2001 Statics, MSE 2001 Materials, MATH 2551 Multivariable Calculus, ME 1670 Graphics, CS 1371 Computing

School of Science and Engineering at Townview – Dallas, TX **August 2022 – May 2026**
3.97 Unweighted GPA | Rank 2 | 1600 SAT | National Merit Scholar | NASA High School Aerospace Scholar

EXPERIENCE

Salvo Dynamics | Design Engineer **August 2026 – Present**
- Turning the aero body of an RC jet built to break the speed of sound into a load-bearing structure in CAD, and laying out the avionics that go inside it
- Designing a 3D printed airframe for testing, plus the assembly jigs used to build the interceptor jets

Hardpoint Robotics | Co-Founder **June 2026 – Present**
- Designing NDAA-compliant development UAVs, owning the airframe: CAD, 3D printed CF-Nylon parts, and carbon fiber structures made with outside vendors
- Designing around sourcing as a hard constraint, since every component has to be compliant and in stock

SWYFT Robotics | Engineering Design Intern **May 2025 – May 2026**
- Designed sprockets, gears, and pulleys for FIRST Tech Challenge and FIRST Robotics Competition teams
- Designed a modular robot cart for FIRST Robotics Competition teams with a detachable programming station

RESEARCH AND ENGINEERING TEAMS

Hypersonics VIP, Georgia Tech | Data and Communications Team **August 2026 – Present**
- Building a sensor package to measure pressure, temperature, and airflow inside a sonic fatigue test facility
- Developing parabolic mirrors to generate a uniform sound field across the facility's test section

Georgia Tech Experimental Rocketry | Externals Team **August 2026 – Present**
- Designing a fin mounting jig that has to locate fins on an already wrapped carbon fiber tube whose exact dimensions are no longer known, and hold fin cant to within a tenth of a degree
- Working out how to measure each fin angle to that tolerance; next step is a mini carbon fiber fin can sent to a third party for dimensioning to validate the jig
- Learning prepreg carbon fiber layup and hands-on composite manufacturing

Yellow Jacket Space Program | Turbomachinery, Engine Development **August 2026 – Present**
- Designing electric pumps and turbopumps so the vehicle can drop the large pressurant tanks a pressure-fed design needs
- Designing and manufacturing impellers, pump housings, and the surrounding hardware
- Team is going for the collegiate liquid propellant altitude record and the Karman line at 100 km

IVALab, Georgia Tech | Undergraduate Researcher **September 2026 – Present**
- Developing robot navigation algorithms for pedestrian dense environments, mixing learned and classical planners and benchmarking them in the Arena simulator

FIRST Tech Challenge Team 7172 | Captain and Lead Designer **September 2021 – May 2026**
- Lead designer on four robots; 2026 World Championship Finalist Alliance, 19 event wins and 63 awards over five seasons
- Replaced an unreliable game piece transfer with one 5-DoF arm doing both intake and scoring, using worm gearboxes and torsion springs to offload the pivot motor; Worlds Division Finalist and Innovate Award

PROJECTS

Beaver Works Summer Institute at MIT | UAS-SAR Track **June 2024 – August 2024**
- Built a synthetic aperture radar from a PulsON 440 on a DJI F550 UAV and wrote the NumPy back projection, which resolved four soda cans next to two much brighter retroreflectors

Pickleball Ball Boy UAV | Personal Project **May 2025 – Present**
- Building a UAV to catch practice serves, testing the perception and guidance stack first on a Jetson Orin Nano holonomic base
- Split the drag model into horizontal and vertical coefficients, bringing landing prediction inside a 20 cm zone

SKILLS

CAD: SolidWorks, Fusion 360, Onshape, Autodesk Inventor
Manufacturing: 3D printing (CF-Nylon), fiber laser cutting, extrusion, carbon fiber vendor work
Programming: Python (NumPy, Matplotlib, Pandas, TensorFlow), MATLAB, JavaScript, SQL, Git
Hardware: NVIDIA Jetson, Raspberry Pi, Arduino, REV Control Hub
Certifications: FAA Part 107 Remote Pilot, Autodesk Certified User (Inventor)